

# Vishal Verma

Junior Data Scientist | Machine Learning Engineer

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## PROFILE SUMMARY

Data Scientist with a strong foundation in Python, SQL, Machine Learning, and Scikit-learn. Skilled in data cleaning, exploratory data analysis, feature engineering, and building end-to-end ML pipelines from statistical validation to model explainability to generate reliable, decision-ready insights. Eager to collaborate on meaningful, analytical work in fast-paced environments.

## SKILLS

**Programming Languages:** Python, SQL, JavaScript (ES6+), Java, C

**Data Science & Machine Learning:** NumPy, Pandas, Scikit-learn, XGBoost, TensorFlow, PyTorch

**Data Visualization:** Matplotlib, Seaborn, MS Excel

**Databases:** MySQL, MongoDB

**Tools & Platforms:** Jupyter Notebook, Git, GitHub, Visual Studio Code, Postman, IntelliJ IDEA

**Analytical Abilities:** Data Cleaning, Exploratory Data Analysis (EDA), Feature Engineering, Hypothesis Testing, Statistical Analysis, Model Evaluation, Predictive Modeling, Insight Generation

## PROJECTS

**UPIShield – UPI Fraud Detection & Risk Intelligence Platform** | Python, MySQL, *Scikit-learn* || [code](#) June – 2026

- Engineered a MySQL fraud database (250K records, indexed views) and ran EDA across 17 attributes, deriving 8 risk features powering 22 business-intelligence SQL queries.
- Built a leakage-safe ML pipeline (ColumnTransformer, SMOTE, Stratified K-Fold CV) to counter 0.19% class imbalance, benchmarking Logistic Regression, Random Forest, and tuned XGBoost via PR-AUC.
- Diagnosed insignificant feature-to-target correlation via chi-square testing, then validated pipeline integrity with a synthetic-pattern experiment — lifting ROC-AUC from 0.51 to 0.74 (SHAP-confirmed).

**LoanSense – Intelligent Loan Approval System** | *Python, XGBoost, Scikit-learn, Streamlit* || [live](#) | [code](#) May – 2026

- Trained and benchmarked 7 classification algorithms (XGBoost, Random Forest, SVM, Logistic Regression, Decision Tree, Naive Bayes, KNN) on 950 records across 20 financial features.
- Achieved 96.3% accuracy and a 0.943 F1-score with XGBoost - best among all seven models - on a 69/31 imbalanced test set via stratified splitting and label/one-hot encoding.
- Deployed a serialized XGBoost pipeline (Joblib) as a Streamlit Community Cloud app, delivering instant loan-eligibility predictions with end-to-end preprocessing.

**SmartCart – Customer Segmentation System** | *Python, Pandas, Scikit-learn, PCA* || [code](#) May – 2026

- Built an unsupervised ML pipeline on 2,240 e-commerce customers using K-Means and Agglomerative Clustering (Ward linkage), reducing 22 features to 3 PCA components.
- Designed a preprocessing pipeline - median imputation, outlier removal, OneHotEncoder, StandardScaler - validating cluster count via Silhouette Score and Elbow Method.
- Profiled four behavioural customer segments, uncovering a VIP cluster with 32% campaign response rate (4 times the baseline), enabling targeted, data-driven marketing.

## EDUCATION

S.D. College of Engineering and Technology (AKTU)

Muzaffarnagar, UP, India

**Bachelor of Technology** in Computer Science and Engineering

2021 – 2025

- Relevant Coursework: Machine Learning, Probability & Statistics, Deep Learning, Data Structures & Algorithms, Database Management Systems, Artificial Intelligence.

## CERTIFICATIONS

- Certified Web Development Course – Apna College (2025) ([link](#))
- C Programming – IIT Kanpur (2021) ([link](#))